

# OSF Amendment 03 — Convention component is an empirical calendar-slab basis (grid-quantisation reframe; no reign tier)

**title:** OSF Amendment 03 — Convention component is an empirical calendar-slab basis (grid-quantisation reframe; no reign tier)

**amendment-number:** 03

**status:** DRAFT (complete) — §A5.5 filled with the full-grid PASS (2026-06-08); awaiting Shawn’s review and lodgement (not yet lodged)

**date-drafted:** 2026-06-07

**scope:** Redefine the H2.1 mixture’s editorial-convention component (p\_conv) as an empirical calendar-slab basis with NO reign tier; reclassify reigns/dynasties/datable events as genuine-but-aoristic; reframe the artefact as grid-quantisation of genuine-but-coarse evidence onto the BC/AD calendar lattice; ride decadal/quarter-century brackets as a sensitivity band. Gated on a fresh recovery re-validation.

**preregistration:** <https://osf.io/uycs6/> (lodged 2026-05-20; embargoed)

**lodged-version:** git tag osf-lodgement-2026-05-20 (<https://github.com/saross/inscriptions/tree/osf-lodgement-2026-05-20>)

**filed-under:** preregistration §7 / contingency clause (preregistration-draft.md line 423): substantive methodology changes after lodgement are filed as an OSF amendment before implementation.

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**gate:** BINDING — the H2.1 temporal-mixture production fit must not run until this amendment is lodged. The cross-sectional track (H3a/H3c/SR1, Amendments 01/02) uses date-window counts, not mixture output, and is NOT gated by this amendment.

**relationship:** Independent of Amendment 01 (two-measure framework, lodged 2026-06-04) and Amendment 02 (Latin-primary frame, lodged 2026-06-06). Both used date-window counts; this one concerns only the temporal mixture’s convention component. Separable lodgement.

**skeleton-note:** §A5.5 filled 2026-06-08 from the full-grid PASS (B = 96.4 %; FULL-GRID-REPORT.md). justification.txt + summary-addendum generated this session; PDF pending. Awaiting Shawn’s review + lodgement.

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## OSF Amendment 03 — Convention component is an empirical calendar-slab basis (grid-quantisation reframe; no reign tier)

### Plain-language summary

Roman inscriptions are mostly dated by convention — “2nd century AD”, “Hadrianic” — rather than to a year. Our method separates that conventional dating fog from the genuine timeline of inscribing. The lodged protocol modelled the convention component as three hand-picked interval types (centuries, half-centuries, and imperial-reign spans). On building the actual dictionary from the corpus, two problems appeared: a large block of the conventional datings are *multi-century* slabs (“[301,500]”) that fit none of the three types, and reign spans turned out to belong on the *genuine* side, not the convention side. This amendment makes three connected changes:

1. **The convention component becomes an empirical, data-built basis** of calendar slab-types (sub-century, century, multi-century), with **no reign tier**. It is built directly from the corpus’s own round-number datings rather than from a hand-picked dictionary.
2. **Reigns, dynasties, and datable events are reclassified as genuine-but-aoristic.** “Flavian” carries real historical information that “the later first century” does not; such dates are genuine signal with wide uncertainty, not editorial convention.

3. **We reframe what “convention” is.** It is not information-free editorial rounding; it is genuine-but-coarse evidence *quantised onto the BC/AD calendar grid*. The method removes that grid-snapping artefact (the boundary pile-ups and flat-within-bin shape) from the collective curve; it does not pretend to recover any single inscription’s true off-grid date. Fine brackets (quarter-centuries, decade windows) are grid-snapped but low-distortion, so they ride as a reported sensitivity band rather than a hard classification.

Because this changes the model’s structure, it is re-validated on fresh synthetic data with known answers before any real fit, and the change is lodged before implementation. The technical statement follows.

## A1. Identification and trigger

This is the third amendment to the project’s preregistration (Open Science Framework, OSF, record [osf.io/uycs6/](https://osf.io/uycs6/), lodged 2026-05-20, currently embargoed), filed under the preregistration’s own contingency rule ([preregistration-draft.md](#) line 423). It is a **binding gate**: the H2.1 temporal-mixture production fit does not run until this amendment is lodged.

**Trigger.** The pre-Phase-2 template-dictionary empirical scan (prereg line 202; [runs/2026-06-05-template-dictionary/](#)) — a preregistered prerequisite for the mixture fit — found the lodged curated three-tier convention dictionary empirically inadequate: multi-century slabs are ~31 % of the F1+F3 convention pool and fit none of the three curated tiers (the single most frequent template corpus-wide is the multi-century [301, 500], 8.8 %), while reign templates are only ~2.7 % and, on inspection, belong on the genuine side. This fired Decision 20’s own revisit trigger. The full deliberation is **Decision 38** ([planning/decision-log.md](#)).

## A2. Summary of the change

Three connected changes to the **temporal mixture’s convention component only**; the model’s likelihood, the F1/F3 structural fixes, the learned-weight count (3), and the cross-sectional track are unchanged (§A6).

1. **Convention basis** → **empirical calendar slab-types, no reign tier** (§A5.1).
2. **Reigns / dynasties / datable events** → **genuine-but-aoristic** (§A5.2).
3. **Grid-quantisation reframe of the artefact** (§A5.3); fine brackets as a sensitivity band (§A5.4).

Lodgement is gated on a **fresh recovery re-validation** (§A5.5) because the basis *shapes* change (they now carry a multi-century plateau), even though the learned tier-weight count stays at the recovery-validated 3.

## A3. Rationale

- **The curated three-tier basis is empirically broken.** It has no home for the ~31 % multi-century mass and over-weights reign (~2.7 %). The recovery grid that returned 98.6 % -coverage ran on *synthetic proxy* tier weights over the curated basis; **no real-LIRE three-tier mixture has ever been fit**.
- **Reigns are genuine, not convention.** The family classifier that operationalises the convention pool already holds reign intervals out (as “F2\_Other”), and it even split reigns inconsistently by width-accident ([117,138] Hadrian held out; [161,180] Marcus leaked in). The lodged prereg’s placement of a reign tier *inside* convention contradicts both the classifier and the conceptual reality.
- **Convention is grid-quantisation, not absence of information.** Every recorded date carries evidential anchoring (letterforms, onomastics, formulae, consular dates, find-context — Cooley 2012). What makes a date *conventional* is the arbitrary rounding of that genuine-but-coarse evidence onto the BC/AD calendar lattice. That snapping introduces (i) per-inscription distortion (an off-grid range truncated to a round century and flattened to uniform) and (ii) cross-inscription artificial alignment (many true distributions snap to the same bin, manufacturing the boundary pile-ups the SPA shows). The discriminator between convention and genuine is therefore **grid-snapping** (observable as grid-alignment), not criterion-type (which LIRE’s `raw_dating` does not preserve).

## A4. Relationship to already-observed results (transparency)

This change is **method-structure, decided before any real-data mixture fit exists** — there is no real-corpus `p_conv/p_gen/` result to have steered it. It was triggered by a descriptive *template-frequency scan* (counts of interval shapes), not by any hypothesis-test outcome. The re-validation (§A5.5) runs on synthetic data with known

answers. The change therefore cannot be result-driven in the confirmatory sense; it is a faithfulness correction to the convention model, disclosed and re-validated before implementation.

## A5. Pre-specifications

### A5.1 Convention basis — empirical calendar slab-types (no reign tier)

$p_{\text{conv}} = \text{tier\_weights} \cdot \text{tier\_basis}$ , with  $\text{tier\_weights} \sim \text{Dirichlet}(\text{ones}(3))$  over a **fixed empirical basis** of three calendar-slab tiers — **sub-century** (half-century), **century**, and **multi-century** (1.5- + 2- + 3-century, pooled) — each row the frequency-weighted aoristic SPA of the anchor-stripped F1+F3 calendar population of that width-class. **No reign tier.** The basis is a fixed, unit-independent convention template, built once per language frame (empire and Latin) and shared across that frame’s units; a per-unit basis would absorb genuine temporal signal into  $p_{\text{conv}}$ . Artefact: `runs/2026-06-06-convention-basis-redesign/design.json` (`tier_basis_empirical[_latin]`).

### A5.2 Reigns / dynasties / events — genuine-but-aoristic

Date assignments tied to real historical anchors (reigns, dynasties, datable events) are **genuine-but-aoristic** and are **not** modelled in  $p_{\text{conv}}$ ; their aoristic mass contributes to the observed SPA and is therefore attributed to the genuine component. A curated historical-anchor interval list (`runs/2026-06-06-convention-basis-redesign/historical-anchor-intervals.json`) removes width-accidental reign/event leaks (empirically just [161,180], 0.11 % of the convention pool) from the basis-building population so the split is non-width-accidental. Year-precise  $[t, \tau]$  dates remain genuine (unchanged).

### A5.3 Grid-quantisation reframe (prereg/paper §2)

Reframe the artefact as *genuine-but-coarse evidence quantised onto the BC/AD calendar grid*, retiring the “editorial rounding no information” / “midpoint- spike” description. Redescribe  $p_{\text{gen}}$  as “*the temporal distribution with the calendar-grid quantisation removed*”, with the honest limit stated: the method un-snaps the **collective** (removes the aggregate boundary pile-ups and flat-within-bin shape under the GRW smoothness prior); it does **not** reconstruct any single inscription’s true off-grid latent distribution. This is distinct from radiocarbon SPD (genuine measurement uncertainty, no arbitrary rounding) and likely shared with ceramic typological dating (round-period pinning).

### A5.4 Fine brackets — sensitivity band

Decadal and quarter-century brackets (~4–5 % of the corpus) are grid-snapped (convention side) but **low-distortion** (fine grid), so deconvolving them vs not barely moves the result. They are **excluded from the primary**  $p_{\text{conv}}$  and added back as a reported robustness band (the ceramics stacked-band idiom), not a hard classification.

### A5.5 Recovery re-validation (the gate) — PASS (triage + full grid)

Grid A’s 98.6 % validated the *old* basis shapes and does **not** transfer to a multi-century-bearing basis (a long flat envelope-edge plateau is confusable with genuine quiescence). The re-validation re-generates synthetics from the new empirical basis and runs an  $= 0.95 \times \text{multi-century} \times \text{peaked-genuine stress-triage first}$ , then the full grid.

- **Stage-1 stress-triage: PASS** (2026-06-06). 8 cells, convergence 1.00; at the hardest corner the model recovers to within +0.029 (0.979 vs true 0.95) — the multi-century plateau is attributed to convention, *not* confused for genuine quiescence. Scored under the Amendment 01 §A5.5.1 criterion ( -coverage = a shape-conditioned diagnostic, not a gate; shape + convergence binding).
- **Stage-2 full grid: PASS** (450 cells, 0 failed; sapphire, 2026-06-08). Headline **B = 96.4 %** of in-envelope ( 0.70) cells are clean passes (convergence AND shape), against the 90 % bar; diagnostic A = 97.2 %. The 13 in-envelope non-passes (3 non-converged + 10 shape-misses) concentrate in the peaked shapes (`regnal_cluster`, large-N bimodal); **the multicentury\_heavy tier is NOT a systematic failure** (it clean-passes at the same rate as every other tier), so the §6 plateau-confusion failure mode is **absent at scale**, and is recovered essentially unbiased (mean signed bias +0.005). The operating envelope ( 0.70) is confirmed; shape recovery collapses only at the out-of-envelope = 0.95 stress row. -recovery precision (Bland–Altman

95 % LoA, in-envelope) is  $[-0.12, +0.13]$  pooled — shape-conditioned  $\pm 0.09$  (smooth/flat) to  $\pm 0.18$  (multimodal) — **within Decision 33's  $\pm 0.18$  envelope**, so the lodged shape-conditioned -hedge carries over unchanged. Basis-shift vs the validated Grid A: 96.4 % vs 98.6 % ( $\Delta -2.2$  %; informational, not a regression target). REPORT: `runs/2026-06-06-convention-basis-redesign/revalidation/FULL-GRID-REPORT.md` (auto-tables `inscription-mass/outputs/REPORT.md`; LoA tables `alpha-loa-summary.json`).

## A5.6 Novelty positioning (verified)

The bracket-level convention-vs-genuine deconvolution survives a verified forward-citation pre-emption chain from Crema 2025. **Cite-and-distinguish** the nearest competitor — Tobalina-Pulido & Martín-Rodilla 2026 (10.5334/jcaa.220), a fuzzy-logic framework that *quantifies/propagates* inherited dating uncertainty but does **not** deconvolve convention from genuine spread. Method-level warrant: **Crema 2025** (10.1111/arc.12984). Dating-method authority: **Cooley 2012** (10.1017/cbo9781139020442); current inscription-dating monograph **Hartmann 2025** (10.46771/978-3-96769-729-2).

## A6. What does NOT change

- **The mixture likelihood and the F1/F3 structural fixes** (`build_model_f1_f3: Beta(1,1)` prior; non-centred GRW `p_gen`) are unchanged.
- **The learned tier-weight count stays 3** — the Dirichlet structure the recovery grid validated is structurally identical; only the basis *shapes* change.
- **The envelope (50 BC – AD 350, 5-y/80-bin) and the observation model** (largest-remainder multinomial) are unchanged.
- **The cross-sectional track** (H3a/H3c/SR1; Amendments 01/02) uses date-window counts, not mixture output, and is untouched.
- **Year-precise dates** remain genuine.

## A7. Changes by preregistration section (a standalone addition, not in-place edits)

- **§2 (the artefact description)**. Reframed as grid-quantisation of genuine-but-coarse evidence onto the BC/AD lattice; retire the midpoint-spike/round-period- -no-information description.
- **§3 (the mixture / `p_conv`)**. Convention component = empirical 3-tier calendar-slab basis (no reign tier); reigns/dynasties/events → genuine-but-aoristic; fine brackets → sensitivity band. Supersedes Decision 20's tier typing; refines Decision 37.
- **§3 (validation)**. The H2.1 recovery validation is re-run on the new basis (§A5.5) before the production fit.

## A8. Provenance

- **Decision record**: Decision 38 (`planning/decision-log.md`); supersedes Decision 20's reign tier; refines Decision 37.
- **Basis + anchor list**: `runs/2026-06-06-convention-basis-redesign/` (`design.json`, `historical-anchor-intervals.json`, `outputs/REPORT.md`).
- **Template-dictionary scan**: `runs/2026-06-05-template-dictionary/` (`6d8950f`).
- **Re-validation**: `runs/2026-06-06-convention-basis-redesign/revalidation/` (`spec.md`, `STAGE1-TRIAGE-REPORT.md`; full-grid REPORT pending).
- **Lodged authority**: the original supplementary (git tag `osf-lodgement-2026-05-20`) remains authoritative for all unchanged specifications.
- **Independence**: separable from Amendments 01 and 02 (both date-window-count based); this concerns only the temporal mixture's convention component.
- **Repository-state provenance**: the git lodgement tag for this amendment (to be created at lodgement) is the reproducibility anchor, distinct from any OSF-assigned amendment identifier.